

Ex.No. 13: V2X Latency Comparison

AIM:

To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Objective:

1. To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.
2. To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.
3. To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

Objective 1 Matlab Code:

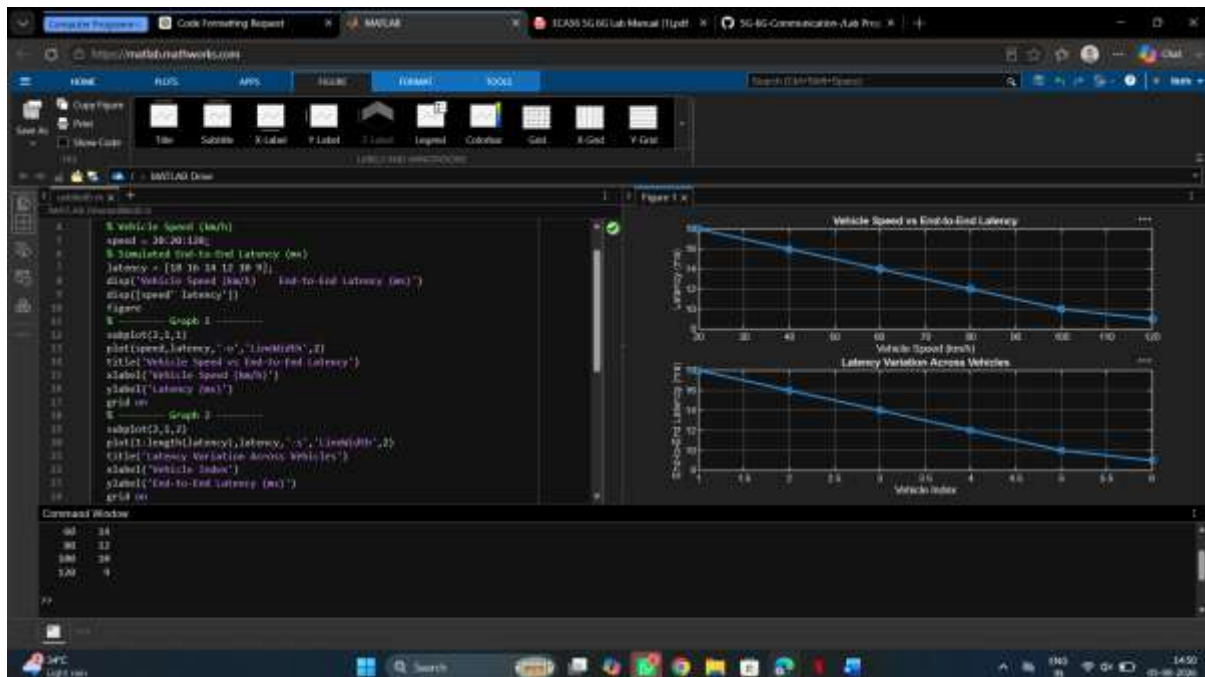
```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h)   End-to-End Latency (ms)')
disp([speed' latency'])
figure;
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed, latency, '-o', 'LineWidth', 2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(latency), latency, '-s', 'LineWidth', 2)
title('Latency Variation Across Vehicles')
```

xlabel('Vehicle Index')

ylabel('End-to-End Latency (ms)')

grid on

OUTPUT:



Objective 2 Matlab Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle Speed (km/h)
```

```
speed = 20:20:120;
```

```
% Simulated Packet Delivery Ratio (%)
```

```
PDR = [99 98 97 95 93 90];
```

```
disp('Vehicle Speed (km/h)   Packet Delivery Ratio (%)')
```

```
disp([speed' PDR'])
```

```
figure;
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(speed, PDR, '-o', 'LineWidth', 2)
```

```
title('Vehicle Speed vs Packet Delivery Ratio')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(1:length(PDR), PDR, '-s', 'LineWidth', 2)
```

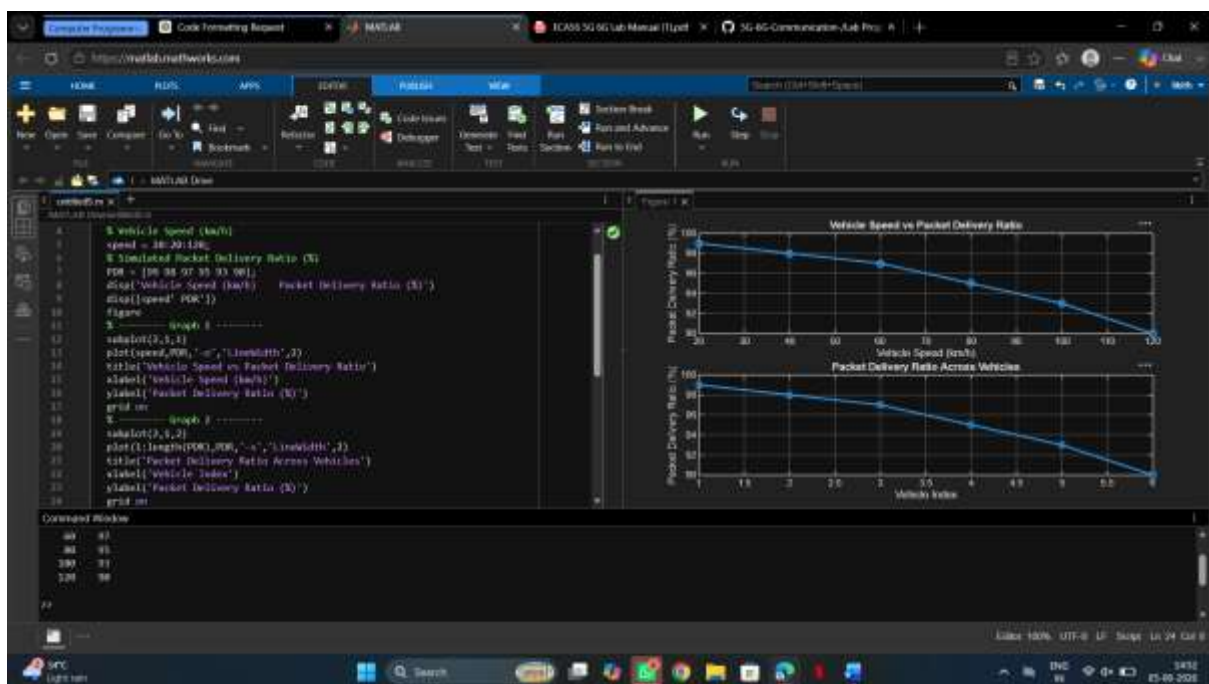
```
title('Packet Delivery Ratio Across Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

OUTPUT :



Objective 3 Matlab Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle IDs
```

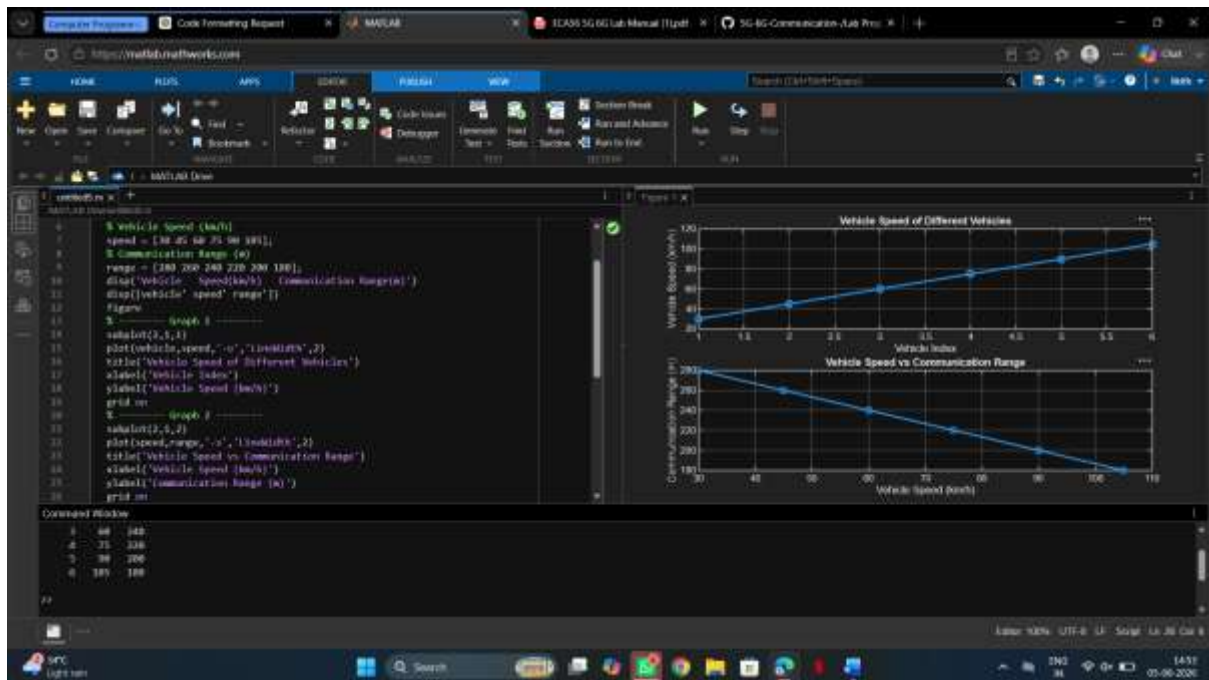
```
vehicle = 1:6;
```

```

% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])
figure;
% ----- Graph 1 -----
subplot(2,1,1)
plot(vehicle, speed, '-o', 'LineWidth', 2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(speed, range, '-s', 'LineWidth', 2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on

```

OUTPUT :



Result:

1. The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.
2. The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.
3. The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.